আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম

الجامعة الإسلامية العالمية شيتاغونغ

International Islamic University Chittagong

Lab Report - 02

COURSE CODE: CSE-2340

COURSE TITLE: SOFTWARE DEVELOPMENT 1

LAB TOPIC:

JAVA PRACTICE PROBLEM

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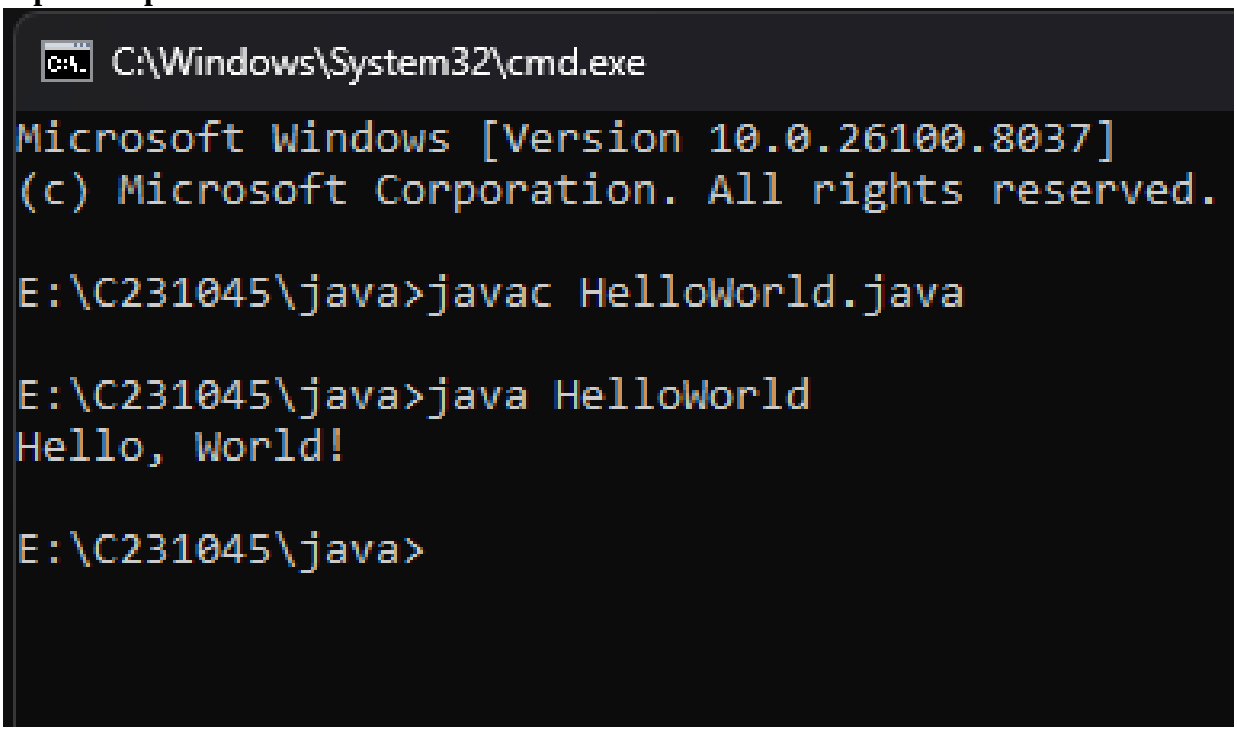
DD-MM-YYYY

Marks:	Signature:

SET-01 (Beginner)

Problem-01:
Print "Hello, World!“

Input-Output Screenshot:



Approach:
I have printed the “Hello, World” message using the println method.

SET-01 (Beginner)

Problem-02:
Take the user's name and greet them. Example, Input: "Your Name" and Output: Hello, "Your Name"!

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac Greeting.java

E:\C231045\java>java Greeting
Enter your name: abdur rahman
Hello, abdur rahman!

E:\C231045\java>
```

Approach:
Firstly, I took the input from the user using the Scanner class of java's utility package. Then, I have printed a hello message with the user's name in the output .

Problem-03:
Check if a number is even or odd **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac EvenOddChecker.java

E:\C231045\java>java EvenOddChecker
Enter a number: 132
132 is an even number

E:\C231045\java>
```

Approach:
In this program, first I took a number as input from the user. Then, using the if-else conditional check if the number is divisible by 2 and the remainder is zero. If the remainder is zero then the number is even otherwise odd.

SET-01 (Beginner)

Problem-04:
Find the maximum of 3 numbers: Take 3 numbers from user input, print the largest.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
icrosoft Windows [Version 10.0.26100.8037]
c) Microsoft Corporation. All rights reserved.

:\C231045\java>javac MaximumOfThree.java

:\C231045\java>java MaximumOfThree
nter first number: 23
nter second number: 10
nter third number: 45
he largest number is: 45

:\C231045\java>
```

Approach:
AT first, I have taken input three numbers from the user with the help of Scanner class. Then, I have used the Math class's max method to calculate the largest between these 3 numbers. I have just to nest one Math.max into another, like this. **Math.max(a, Math.max(b,c))**.

SET-01 (Beginner)

Problem-05:
Simple Calculator (Add, Subtract, Multiply, Divide): Use if-else or switch to perform operations.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
E:\C231045\java>javac SimpleCalculator.java

E:\C231045\java>java SimpleCalculator
Enter first number: 10
Enter second number: 5
Enter operation (+, -, *, /): +
10.0 + 5.0 = 15.0

E:\C231045\java>java SimpleCalculator
Enter first number: 45
Enter second number: 85
Enter operation (+, -, *, /): -
45.0 - 85.0 = -40.0

E:\C231045\java>java SimpleCalculator
Enter first number: 100
Enter second number: 250
Enter operation (+, -, *, /): *
100.0 * 250.0 = 25000.0

E:\C231045\java>java SimpleCalculator
Enter first number: 10
Enter second number: 20
Enter operation (+, -, *, /): /
10.0 / 20.0 = 0.5
```

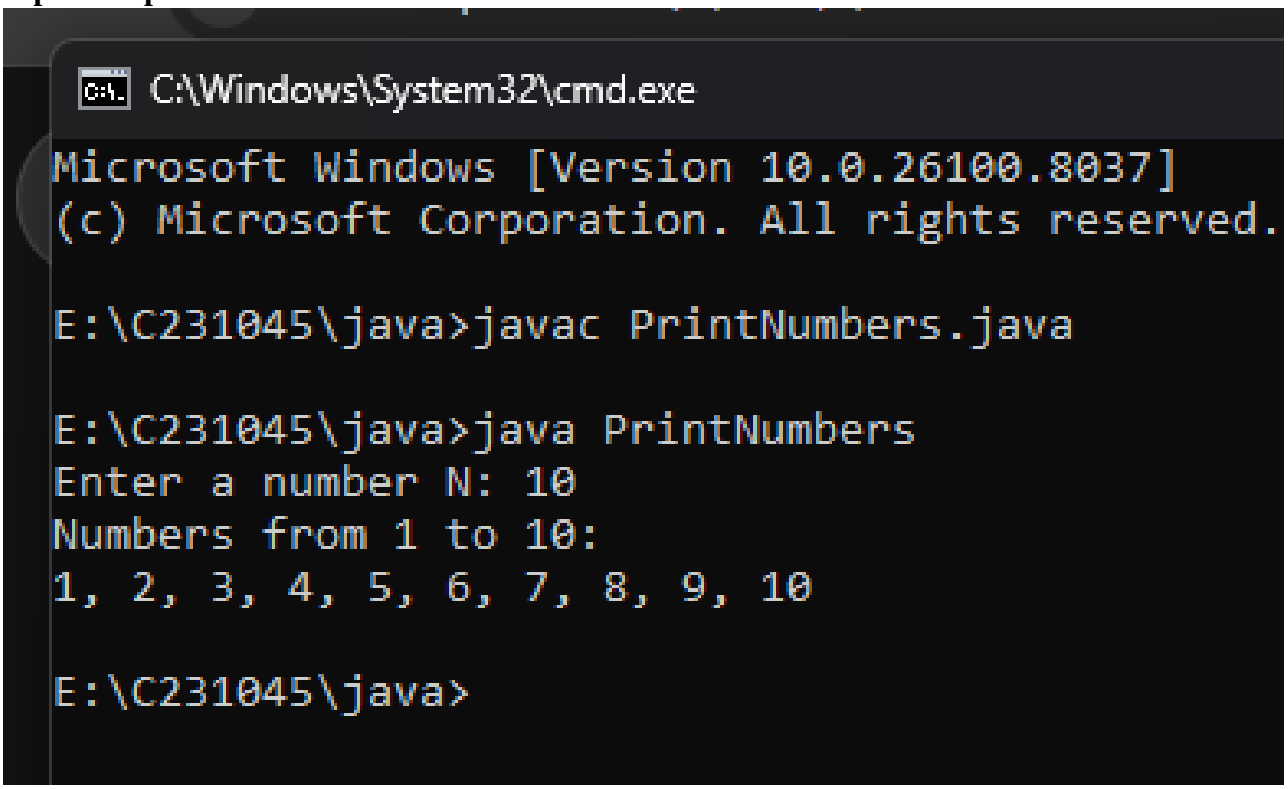
Approach:
First I have taken 3 inputs from the users where 2 are float typed variables and one is a char typed operator for operation sign. Then, I have used a switch case to calculate the result depending on the sign provided by the user. And finally, printed the result.

SET-01 (Beginner)

Problem-06:

Print numbers from 1 to N. Take N from the user input.

Input-Output Screenshot:



Approach:

AT First I have taken the input from the user using the Scanner class and stored the value in an int typed variable. Then I have run a for loop to loop through from 1 to n. I take 10 .

Problem-07:
Print the multiplication table of a number **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac MultiplicationTable.java

E:\C231045\java>java MultiplicationTable
Enter a number: 5
Enter the range (how many multiples): 10

Multiplication Table of 5:
==================================
5 ? 1 = 5
5 ? 2 = 10
5 ? 3 = 15
5 ? 4 = 20
5 ? 5 = 25
5 ? 6 = 30
5 ? 7 = 35
5 ? 8 = 40
5 ? 9 = 45
5 ? 10 = 50
```

Approach:
Here, I have first taken the number input from the user using the Scanner class. Then, I ran a for loop from 1 to 10. Inside the for loop, I have calculated the multiplication table using (n * i) and printed the output in the terminal.

SET-01 (Beginner)

Problem-08:
Calculate the factorial of a number. Example: 5! = 5 × 4 × 3 × 2 × 1 = 120 **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac Factorial.java

E:\C231045\java>java Factorial
Enter a non-negative integer: 5
5! = 5 ? 4 ? 3 ? 2 ? 1 = 120

E:\C231045\java>
```

Approach:
Here, first I have taken the input from the user using the Scanner class and set the value of the factorial variable to 1.

Then, I have run a while loop that runs on the condition of n > 1. Inside that i have multiplied the factorial variable with n in each iteration and decrement n by 1 afterwards.

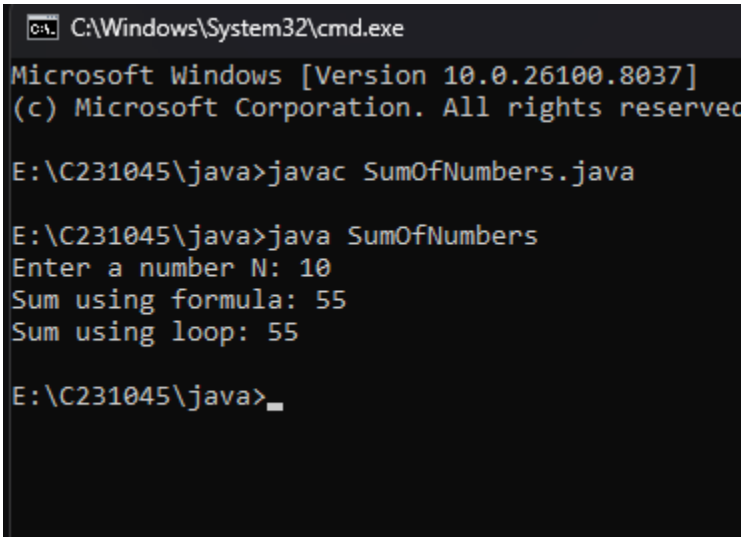
Finally, print the factorial in the last line.

SET-01 (Beginner)

Problem-09:

Sum of all numbers from 1 to N **Input-Output**

Screenshot:



Approach:

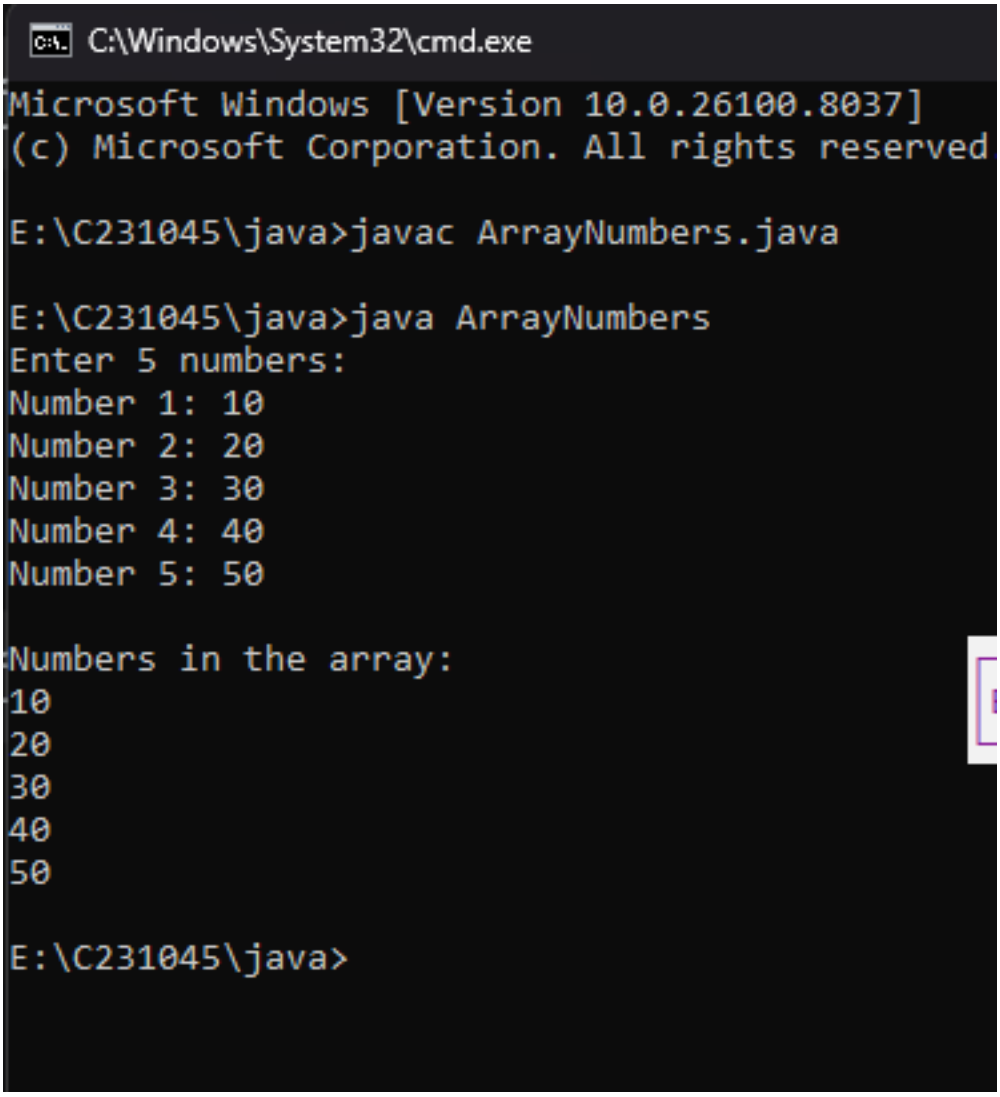
Here, I have taken the input from the user and stored it in the n (int) variable. Then, I have declared another int type variable called sum and there I have written an expression that is used to calculate the sum from 1 to n. The expression is: (n * (n+1))/2

Finally, I have printed the sum in the terminal.

SET-02 (Intermediate)

Problem-1:
Store and print 5 numbers in an array **Input-Output**

Screenshot:



Approach:
Here, I have used the ArrayList to store 5 integers in it using a while loop. Then I have used a for each loop to print the stored values

SET-02 (Intermediate)

Problem-02:
Find the largest number in an array **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac LargestNumber.java

E:\C231045\java>java LargestNumber
Enter the size of array: 5
Enter 5 numbers:
Number 1: 23
Number 2: 45
Number 3: 12
Number 4: 67
Number 5: 34

Largest number in the array: 67

E:\C231045\java>
```

Approach:
Here, at first I have declared an array with 5 integer elements. Then, I have run a for loop to display the elements inside the array.

Then, I have run another for loop that finds the largest element among the integers inside the array, using a condition, arr[i] > max

max = arr[i] .them print large number

SET-02 (Intermediate)

Problem-03:
Find the smallest number in an array **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac SmallestNumber.java

E:\C231045\java>java SmallestNumber
Enter the size of array: 5
Enter 5 numbers:
Number 1: 100
Number 2: 2545
Number 3: 654
Number 4: 65487
Number 5: 021

Smallest number in the array: 21

E:\C231045\java>
```

Approach:
Here, at first I have declared an array with 5 integer elements. Then, I have run a for loop to display the elements inside the array.

Then, I have run another for loop that finds the smallest element among the integers inside the array, using a condition, arr[i] < min

min = arr[i] .and print smallest number

SET-02 (Intermediate)

Problem-04:

Calculate the average of an array **Input-**

Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac AverageArray.java

E:\C231045\java>java AverageArray
Enter the size of array: 5
Enter 5 numbers:
Number 1: 102
Number 2: 20
Number 3: 30
Number 4: 40
Number 5: 50

Sum of array: 242
Average of array: 48.4

E:\C231045\java>
```

Approach:

First, I have declared an array of integers, where the size of the array is 5. Then, I have run a for loop to display all elements in the array. Then I have run another for loop that stores the sum of all the elements of the array in a sum variable.

Then I have created another variable called avg with float type and type casted the sum variable to float and divided it by the length of the array.

Finally, I printed the average.

SET-02 (Intermediate)

Problem-05:
Count even and odd numbers in an array **Input-Output**

Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac EvenOddCount.java

E:\C231045\java>java EvenOddCount
Enter the size of array: 6
Enter 6 numbers:
Number 1: 23
Number 2: 44
Number 3: 17
Number 4: 38
Number 5: 91
Number 6: 42

Even numbers: 3
Odd numbers: 3

E:\C231045\java>
```

Approach:
I have first declared an array of some integers. Then, I have run a for loop to display the elements in the array. Then, another for loop for counting the even and odd numbers in the array. Inside that loop, I have written a condition, like arr[i] % 2 == 0, then it is an even number, else it is an odd number.

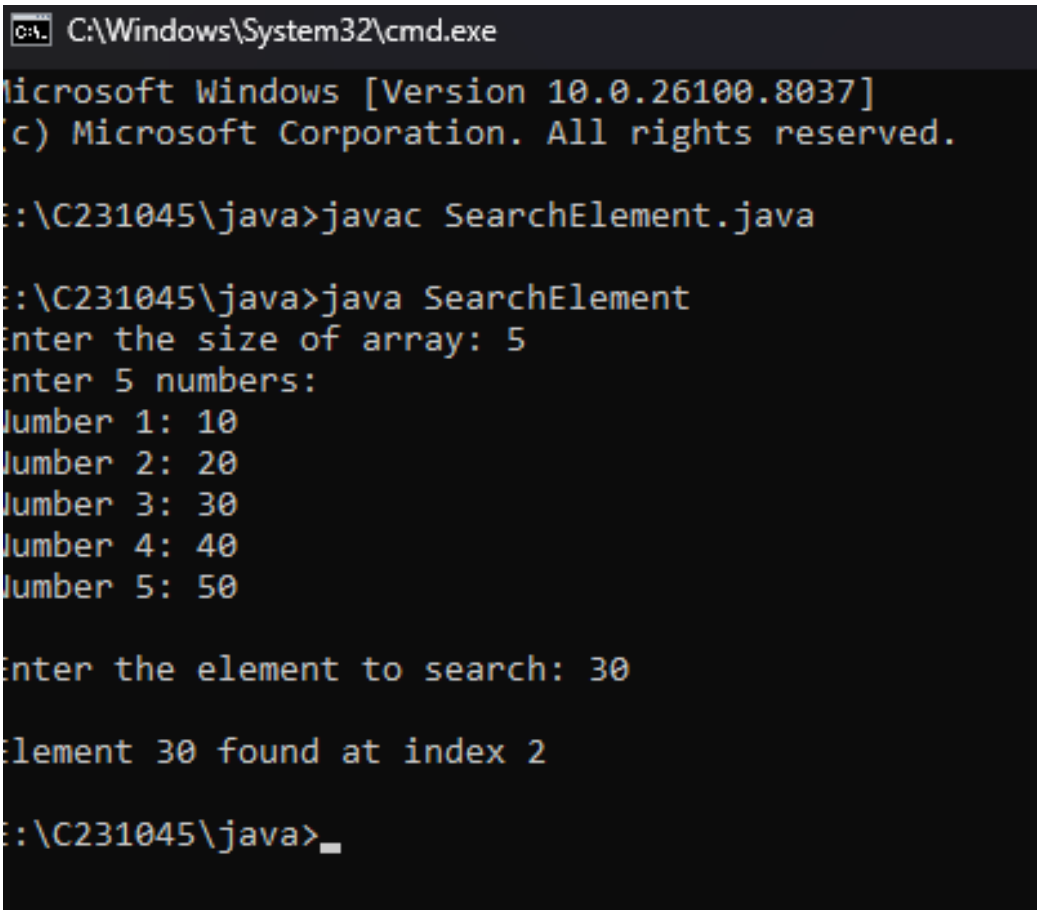
Finally, print the even number and odd number counts.

SET-02 (Intermediate)

Problem-06:

Search for an element in an array **Input-**

Output Screenshot:



Approach:

Here, I have implemented the binary search that works on a sorted array. Here, my array was sorted by default. And it looked something like this, {1,2,3,4,5} and the element I was searching for was 4. So, in binary search, we just make the array length half in each iteration to find out our target element. Here, I did that using a while loop comparing the low and high position of the array.

SET-02 (Intermediate)

Problem-07:
Sort an array (Ascending Order) **Input-**

Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac SortArray.java

E:\C231045\java>java SortArray
Enter the size of array: 6
Enter 6 numbers:
Number 1: 45
Number 2: 23
Number 3: 67
Number 4: 12
Number 5: 89
Number 6: 34

Array in ascending order:
12 23 34 45 67 89
E:\C231045\java>
```

Approach:
Here, I have implemented bubble sort to sort the array in ascending order. I have used two nested for loops. The inner one is the one that actually does the work of swapping the largest element of the occurrence to the end of the array with the help of a condition that checks if one element is larger than its next element.

SET-02 (Intermediate)

Problem-08:
Check if a string is a palindrome **Input-Output**

Screenshot:

```
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac PalindromeCheck.java

E:\C231045\java>java PalindromeCheck
Enter a string: nadan
"nadan" is a palindrome

E:\C231045\java>java PalindromeCheck
Enter a string: hellow
"hellow" is not a palindrome

E:\C231045\java>_
```

Approach:
Here, I have used the two-pointer approach. Where in each iteration inside the loop we close to the center of the string from left and right direction. And in each iteration we also check if the current left value equals the right end character value. If not, we immediately break the loop and say its not a palindrome. Otherwise, it's a palindrome string.

SET-02 (Intermediate)

Problem-09:
Count vowels and consonants in a string **Input-Output**

Screenshot:

```
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac VowelConsonantCount.java

E:\C231045\java>java VowelConsonantCount
Enter a string: hellow world

String: "hellow world"
Vowels: 3
Consonants: 8

E:\C231045\java>_
```

Approach:
Here, I simply run a for loop that goes through each character of the string and checks if it is a vowel or not, if it is then increase the vowels count otherwise increases the consonants counts. And, finally display it in the terminal.

SET-02 (Intermediate)

Problem-10:
Convert a string to uppercase and lowercase

Screenshot:

```
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac StringCaseConverter.java

E:\C231045\java>java StringCaseConverter
Enter a string: hello world

Original string: hello world
Uppercase: HELLO WORLD
Lowercase: hello world

E:\C231045\java>
```

Approach:
Here, I simply run a for uppercase and lowercase letter.

SET-03 (Descriptive Problems)

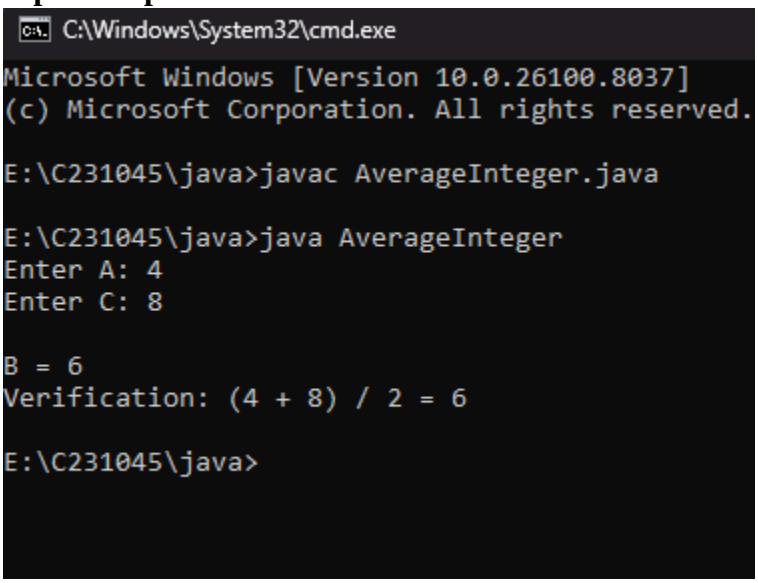
Problem-1:

You are given 2 integers - A and C. You need to find if there exists any integer B

that meets the following condition -B must be an integer

-B is the average of A and C

Input-Output Screenshot:



Approach:

Here, I have first taken two integers A and C as input using the Scanner class of Java utils. Then, I have used a condition block to check if there is a remainder in the average of A and C. If no remainder, then print B exists and it is the average of A and C. Otherwise, print B doesn't exist as none of the conditions met.

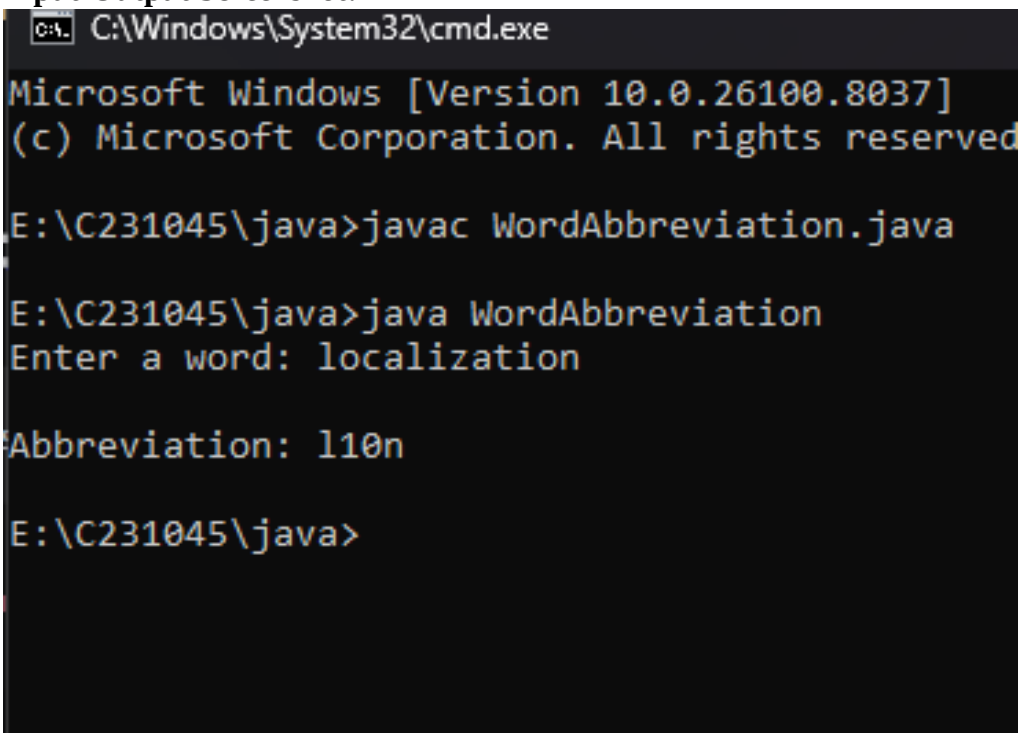
SET-03 (Descriptive Problems)

Problem-2:

Sometimes, some words like "localization" or "internationalization" are so long that writing them many times in one text is quite tiresome. Let's consider a word too long, if its length is strictly more than 10 characters. All too long words should be replaced with a special abbreviation.

This abbreviation is made like this: we write the first and last letters of a word, and between them we write the number of letters between them. That number is in decimal and doesn't contain any leading zeros. Thus, "localization" will be spelt as "l10n", and "internationalization" will be spelt as "i18n". You are suggested to automate the process of changing words with abbreviations. At that, all too long words should be replaced by the abbreviation and the words that are not too long should not undergo any changes.

Input-Output Screenshot:



Approach:

Here, I have first taken the string input from the user using the Scanner class. After that, I have run a condition block based on the length of the string, if it is more than 10 then, I have stored the first and last character of the string into two separate variables. Then, I have stored the count of words between first and last in an integer variable. Then, created a final string concatenating the values of the firstChar + totalCharsInBet + lastChar. Then, printed that final string. If the string is less than 10, I have just printed the original string.

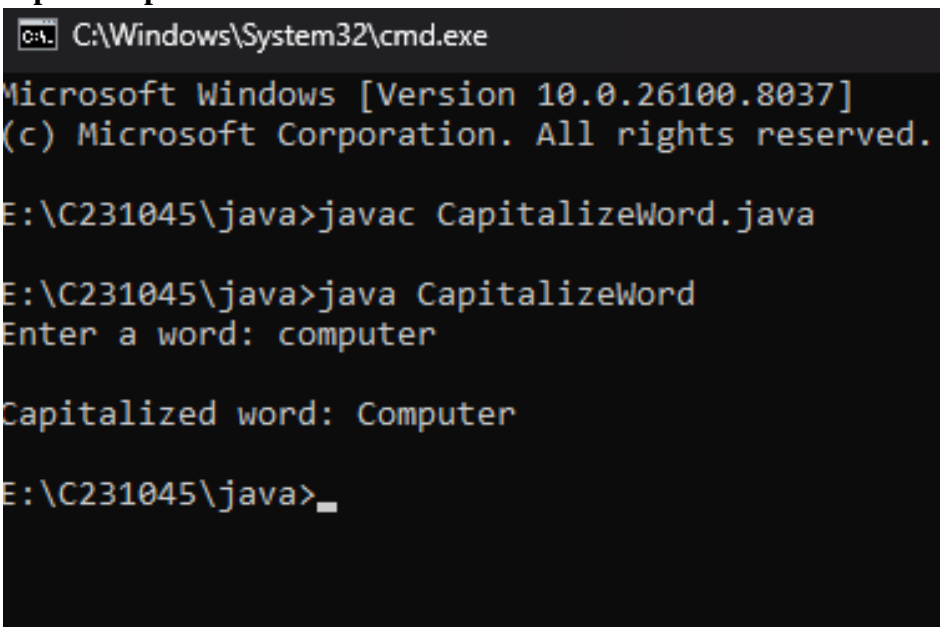
SET-03 (Descriptive Problems)

Problem-3:

Capitalization is writing a word with its first letter as a capital letter. Your task is to capitalize the given word.

Note that during capitalization, all the letters except the first one remain unchanged.

Input-Output Screenshot:



Approach:

So, I have first taken the string input from the user using the Scanner class of Java utils. Then, I have taken out the first character of the string using the substring method and capitalized it with the toUpperCase method and stored it in a firstChar variable of string type. And then, I have created another string where I used again the substring method to take out the rest of the characters except the first character and stored them there. Finally, created a finalString and combined both these strings and printed the finalString on the terminal.

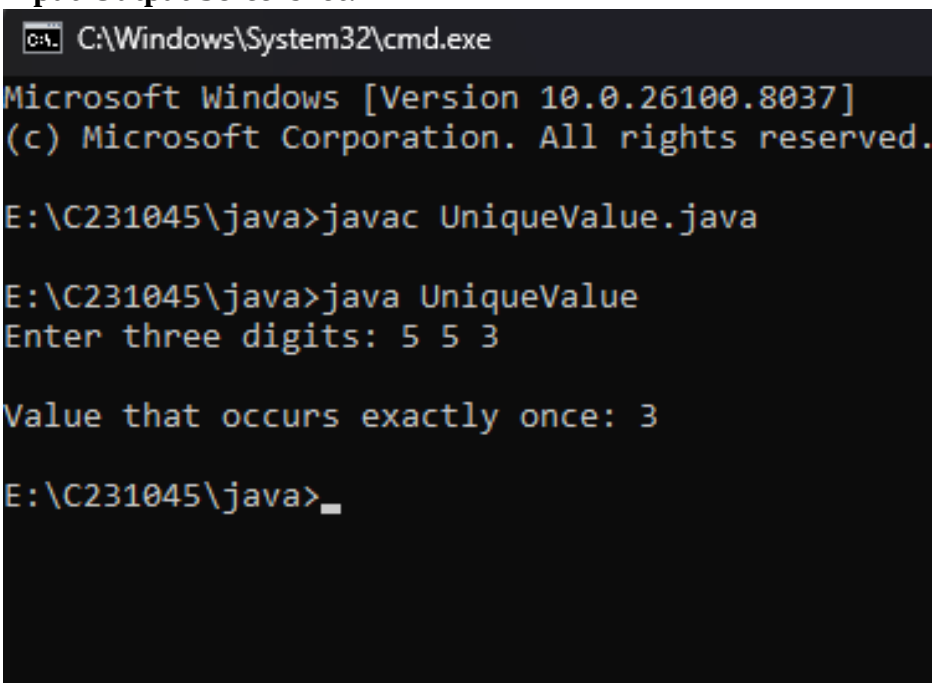
SET-03 (Descriptive Problems)

Problem-4:

You are given three digits, a, b, and c. Two of them are equal, but the third one is different from the other two.

Find the value that occurs exactly once

Input-Output Screenshot:



Approach:

At first, I took three integers as input and stored them in a, b and c. Then, I have run an if-else condition block that

SET-03 (Descriptive Problems)

Problem-5:

You are given 3 numbers A, B, and C. Determine whether the average of A and B is strictly greater than C or not?

NOTE: Average of A and B is defined as (A+B)/2. For example, the average of 5 and 9 is 7, and the average of 5 and 8 is 6.5.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac AverageComparison.java

E:\C231045\java>java AverageComparison
Enter A: 10
Enter B: 8
Enter C: 7

Yes, average (9.0) is strictly greater than C (7.0)

E:\C231045\java>java AverageComparison
Enter A: 5
Enter B: 9
Enter C: 8

No, average (7.0) is not strictly greater than C (8.0)

E:\C231045\java>java AverageComparison
Enter A: 6
Enter B: 8
Enter C: 7

No, average (7.0) is not strictly greater than C (7.0)

E:\C231045\java>
```

Approach:

Here, first I have taken three integers input from the user and stored them in A, B and C respectively. Then, I have stored the average of the A and B in a float type variable called avg. While calculating the average, I have type-casted A inside the operation. So, we get the floating value as well. Then, I just ran a simple if-else condition to check if the (avg > C) .

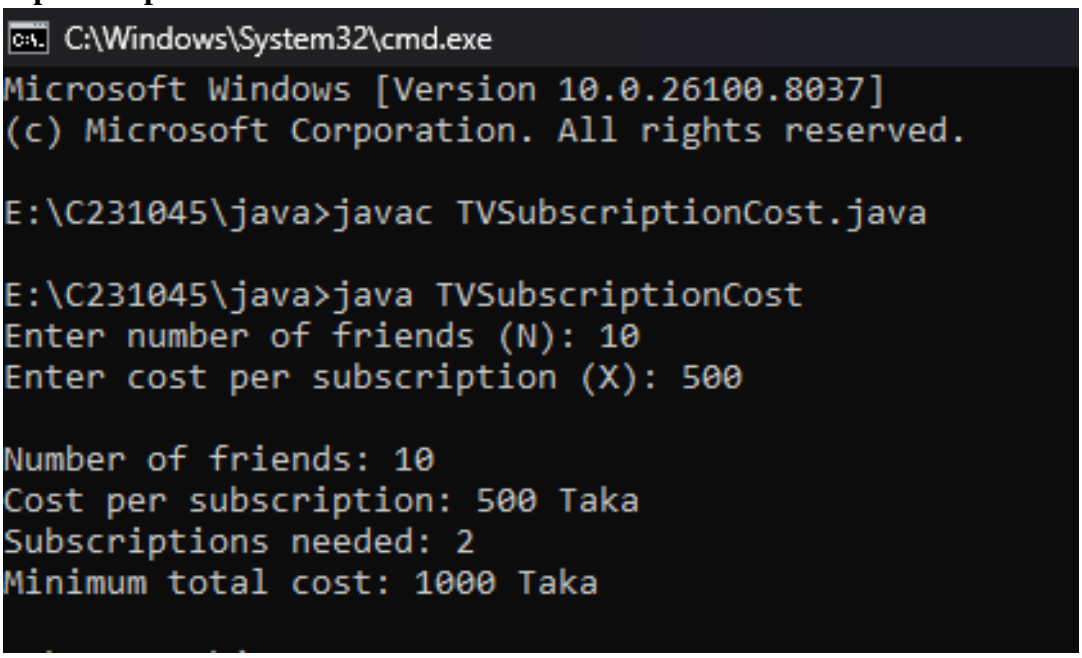
If yes, then we show the average is greater than C. Otherwise, the average is not greater than C.

SET-03 (Descriptive Problems)

Problem-6:

A new TV streaming service, IIUC-TV, was recently launched in IIUC Hall. A group of N friends in IIUC wants to buy IIUC-TV subscriptions. We know that 6 people can share a single IIUC-TV subscription, IIUC. Also, the cost of one IIUC-TV subscription is X Taka. Determine the minimum total cost that the group of N friends will incur to use IIUC-TV, IIUC.

Input-Output Screenshot:



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac TVSubscriptionCost.java

E:\C231045\java>java TVSubscriptionCost
Enter number of friends (N): 10
Enter cost per subscription (X): 500

Number of friends: 10
Cost per subscription: 500 Taka
Subscriptions needed: 2
Minimum total cost: 1000 Taka
```

Approach:

So, here first I have taken the number of friends and cost per subscription from the user as input. Then, I have calculated the number of subscriptions that will be needed for that group of friends, using n / 6, as per subscription 6 people are allowed. Then, I calculated the total cost of the subscriptions for that group using Math.ceil(totalSubs) * x.

But, one important thing was I had to type-cast the Math.ceil to int as Math.ceil returns double and storing it in an integer variable like totalCost doesn't make sense. Finally, printed the total cost.

SET-03 (Descriptive Problems)

Problem-7:

A student has to submit 3 assignments before 10 pm, and he starts to do the assignments at X pm. Each assignment takes him 1 hour to complete. Can you tell whether he'll be able to complete all assignments on time?

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac AssignmentDeadline.java

E:\C231045\java>java AssignmentDeadline
Enter start time (X pm): 6

Yes, he will complete all assignments on time
Completion time: 9 pm
Deadline: 10 pm

E:\C231045\java>java AssignmentDeadline
Enter start time (X pm): 8

No, he will not complete all assignments on time
Completion time: 11 pm
Deadline: 10 pm
Late by: 1 hour(s)

E:\C231045\java>java AssignmentDeadline
Enter start time (X pm): 7

Yes, he will complete all assignments on time
Completion time: 10 pm
Deadline: 10 pm
```

Approach:

Here, I have first taken the starting time from the user and stored it in an integer type variable. Then, I calculated the finishTime based on the start time by adding 3 to the start time as it takes one hour to finish 1 assignment and the student has to finish 3. Then, if the finishTime <= 10, print he can finish on time. Otherwise, he can't.

SET-03 (Descriptive Problems)

Problem-8:

In IIUC, there are X sections, and each section has Y students. The semester-end results are in, and a total of Z students passed the exams. Assuming that all students appeared for the exams, find whether the number of students who passed was strictly greater than 50%.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Enter number of sections (X): 5
Enter students per section (Y): 40
Enter number of students passed (Z): 120

Yes, passed students are strictly greater than 50%
Total students: 200
Passed students: 120
Pass percentage: 60.00%

E:\C231045\java>java PassPercentage
Enter number of sections (X): 4
Enter students per section (Y): 25
Enter number of students passed (Z): 50

No, passed students are not strictly greater than 50%
Total students: 100
Passed students: 50
Pass percentage: 50.00%

E:\C231045\java>java PassPercentage
Enter number of sections (X): 3
Enter students per section (Y): 30
Enter number of students passed (Z): 49

Yes, passed students are strictly greater than 50%
Total students: 90
Passed students: 49
Pass percentage: 54.44%
```

Approach:

First I have taken the number of sections and stored it in X, then I have taken the number of students in each section and stored it in Y, then, the total number of students who had passed in the exam and stored it in z. Then, I have calculated the total number of students using x * y. Then, I calculated the percentage of passing students. Percentage = (total number of students passed / total number of students) * 100

Finally, I have checked the condition if the percentage is greater than 50 or not. If greater than 50, then print the percentage is greater than 50 for this exam. Otherwise, the percentage is not greater than 50 for this exam.

SET-03 (Descriptive Problems)

Problem-9:

Recently you joined a new company. At this company, employees must work X hours each day from Saturday to Wednesday. Also, in this company, Wednesday is called Chill Day; employees only have to work for Y hours (Y<X) on Wednesday.

Determine the total number of working hours in one week.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac WeeklyWorkingHours.java

E:\C231045\java>java WeeklyWorkingHours
Enter regular working hours (X): 8
Enter Wednesday working hours (Y): 5

=== Weekly Working Hours Summary ===
Saturday to Tuesday: 4 days ? 8 hours = 32 hours
Wednesday (Chill Day): 1 day ? 5 hours = 5 hours
Total working hours: 37 hours

E:\C231045\java>java WeeklyWorkingHours
Enter regular working hours (X): 9
Enter Wednesday working hours (Y): 6

=== Weekly Working Hours Summary ===
Saturday to Tuesday: 4 days ? 9 hours = 36 hours
Wednesday (Chill Day): 1 day ? 6 hours = 6 hours
Total working hours: 42 hours

E:\C231045\java>_
```

Approach:

First, I have taken the input from the users on how many hours he works each day (x), then, how many hours work in Wednesday (y). After getting the input from the user, I have calculated the total hours the user works per week from Saturday-Wednesday using the formula,

Total Hours = (Working hours per day * 4) + Working hours on Wednesday

Finally, printed the totalHours in the terminal.

SET-03 (Descriptive Problems)

Problem-10:

An IIUCian has three water bottles. At any point, if at least two of them are empty, he will fill them up. But if at most one bottle is empty, he will wait and not fill them up now.

You are given three integers: B1, B2, and B3. If B1=1, then the first bottle is full. If B1=0, then the first bottle is empty. Similarly, B2 denotes whether the second bottle is full or empty, and B3 denotes it for the third bottle. Output "Water filling time", if he has to fill the bottles now. If not, output "Not now".

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac WaterFilling.java

E:\C231045\java>java WaterFilling
Enter B1 (1 for full, 0 for empty): 0
Enter B2 (1 for full, 0 for empty): 0
Enter B3 (1 for full, 0 for empty): 1

Water filling time
Empty bottles: 2

E:\C231045\java>java WaterFilling
Enter B1 (1 for full, 0 for empty): 1
Enter B2 (1 for full, 0 for empty): 0
Enter B3 (1 for full, 0 for empty): 1

Not now
Empty bottles: 1

E:\C231045\java>_
```

Approach:

Here, first I took the status of B1, B2 and B3 as input. Then, I have run a condition checking if B1+B2 == 0, then its Water filling time. If B2 + B3 == 0, then, water filling time, if B1+B3 == 0, then its water filling time. Otherwise, print “Not now”

We wrote the conditions like this, to avoid writing the same logical operator based conditions again and again. This was much simpler and works perfectly with our problem.

SET-03 (Descriptive Problems)

Problem-11:

You know that 1 kg of pulp can be used to make 1000 pages and 1 notebook consists of 100 pages. Suppose a notebook factory receives N kg of pulp. How many notebooks can be made from that?

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac NotebookProduction.java

E:\C231045\java>java NotebookProduction
Enter amount of pulp (N kg): 5

=== Production Summary ===
Pulp: 5 kg
Total pages produced: 5000 pages
Notebooks that can be made: 50 notebooks

E:\C231045\java>
```

Approach:

Here, I first took the number of kgs of pulp and stored it in n (int). Then, I have calculated the total pages count by multiplying n to 1000. As, from 1 kg pulp we get 1000 pages. Then, I calculated the number of notebooks that could be made out of those pages. By dividing the total pages by 100 we get the number of total books as 100 pages are in 1 notebook.

Finally, I printed the total number of notebooks that can be made out of n kg pulp.

SET-03 (Descriptive Problems)

Problem-12:

There are N children, and an IIUCian wants to give each of them 1 packet of candy. He already has X candies. To buy the rest, he visits a candy shop. In the shop, packets containing exactly 4 candies are available.

Determine the minimum number of candy packets he must buy so that he can give 1 candy packet to each of the N children.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe - java CandyPackets
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac CandyPackets.java

E:\C231045\java>java CandyPackets
Enter number of children (N): 10
Enter candies already have (X): 3

=== Summary ===
Children: 10
Candies already have: 3
Candies needed: 7
Packets to buy (4 candies each): 2

E:\C231045\java>java CandyPackets
Enter number of children (N): 8
Enter candies already have (X): 10

=== Summary ===
Children: 8
Candies already have: 10
Candies needed: -2
Packets to buy (4 candies each): 0

E:\C231045\java>java CandyPackets
Enter number of children (N): 03000000000000
```

Approach:

Here, First I have taken the number of children and number of already available candies from the user as input.

Here, each packet of candies has 4 candies in it and the user must be able to divide 1 packet to each child.

So, I wrote an algorithm that gives the total number of packets of candies the user must buy. It looks something like this -

SET-03 (Descriptive Problems)

Problem-13:

An IIUCian has reached the finals of the Annual Inter-university Declamation contest. For the finals, students were asked to prepare 10 topics. However, he was only able to prepare three topics, numbered A, B, and C; he is totally blank about the other topics. This means he can only win the contest if he gets to speak on topics A, B, or C. On the contest day, he gets topic X.

Determine whether he has any chances of winning the competition. Print "Yes" if it is possible to win the contest, else print "No".

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.26100.8037]
(c) Microsoft Corporation. All rights reserved.

E:\C231045\java>javac WinningTopic.java

E:\C231045\java>java WinningTopic
Enter prepared topic A: 3
Enter prepared topic B: 7
Enter prepared topic C: 9
Enter contest topic X: 7

Yes
Topic 7 is prepared!

E:\C231045\java>java WinningTopic
Enter prepared topic A: 1
Enter prepared topic B: 4
Enter prepared topic C: 6
Enter contest topic X: 8

No
Topic 8 is not prepared

E:\C231045\java>
```

Approach:

Here, I have first taken the 3 topics count from the user and stored it respectively in A, B and C. Then, I have taken the count of the topic that came on the contest day and stored it in x. Then, I have just ran a conditional block that just checks if x equals any of A,B and C. If it matches any of these, then print “Yes”. Otherwise, print “No” as he has no chance of winning the competition.

SET-03 (Descriptive Problems)

Problem-14:

There are 4 companies in the markets of Chittagong, A, B, C, and D. This year,

a. Company A made a profit of P lakh,

b. Company B made a profit of Q lakh,

c. Company C made a profit of R lakh,

d. Company D made a profit of S lakh.

There is said to be a monopoly in the market if the profit made by one company is strictly greater than the sum of profits made by all other companies. Determine whether there is a monopoly in the market.

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Enter profit of Company B (Q lakh): 25
Enter profit of Company C (R lakh): 20
Enter profit of Company D (S lakh): 25

=== Market Analysis ===
Company A: 30 lakh
Company B: 25 lakh
Company C: 20 lakh
Company D: 25 lakh

No monopoly in the market
No company's profit is strictly greater than the sum of others

E:\C231045\java>java MonopolyCheck
Enter profit of Company A (P lakh): 15
Enter profit of Company B (Q lakh): 40
Enter profit of Company C (R lakh): 10
Enter profit of Company D (S lakh): 10

=== Market Analysis ===
Company A: 15 lakh
Company B: 40 lakh
Company C: 10 lakh
Company D: 10 lakh

Monopoly exists!
Company B has monopoly
Its profit is greater than the sum of all other companies
```

SET-03 (Descriptive Problems)

Problem-15:

Approach:
Here, first I have taken the profits of A,B,C and D and stored the values in P, Q, R and S. Then, I have created a function that has 3 parameters x,y and z and it returns the sum of x, y and z. Then, I have run a conditional block checking -

if P is greater than sum(Q,R,S), if it is then printed there is monopoly in the market.

If Q is greater than sum(P,R,S) or if it is then printed there is monopoly in the market.

If R is greater than sum(P,Q,S) or if it is then printed there is monopoly in the market.

If S is greater than sum(P,Q,R) or if it is then printed there is monopoly in the market.

If none of the conditions above are met, then say there is no monopoly in the market.

SET-03 (Descriptive Problems)

Problem-15:

There are three people sitting in a room: Dean Sir, Chairman Sir, and Coordinator Sir. They need to decide on the temperature to set on the air conditioner. Everyone has a demand each:

a. Alice wants the temperature to be at least A degrees.

b. Bob wants the temperature to be at most B degrees.

c. Charlie wants the temperature to be at least C degrees.

Can they all agree on some temperature, or not?

Input-Output Screenshot:

```
C:\Windows\System32\cmd.exe
Enter Dean Sir's minimum temperature (A): 20
Enter Chairman Sir's maximum temperature (B): 30
Enter Coordinator Sir's minimum temperature (C): 25

Yes, they can agree!
Temperature range: 25° to 30°
Suggested temperature: 25°

E:\C231045\java>java TemperatureAgreement
Enter Dean Sir's minimum temperature (A): 25
Enter Chairman Sir's maximum temperature (B): 28
Enter Coordinator Sir's minimum temperature (C): 30

No, they cannot agree!
Dean Sir wants: ≥ 25°
Coordinator Sir wants: ≥ 30°
Combined minimum requirement: ≥ 30°
Chairman Sir's maximum: ≤ 28°
No temperature satisfies both conditions

E:\C231045\java>java TemperatureAgreement
Enter Dean Sir's minimum temperature (A): 22
Enter Chairman Sir's maximum temperature (B): 25
Enter Coordinator Sir's minimum temperature (C): 22

Yes, they can agree!
Temperature range: 22° to 25°
Suggested temperature: 22°
```

Approach:

Here, I first took three temperatures desired by the three people and stored it in A,B and C. Then, I have found out the max temperature required by using Math.max(A,C) and stored it in a highestMinReq variable. Then, check if highestMinReq is less than equal B, if yes, print They all can agree on the same temperature. Otherwise, they can't.

THANK YOU